Nicholas Garza

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Experience

Microsoft
Redmond, WA

Software Engineer II - Xbox Graphics August 2020 - Present

- Designed and deployed a web-based performance dashboard to capture FPS regressions on driver and chipset changes across handheld and desktop devices, used demonstrate driver optimizations in leadership syncs
- Automated the next generation Xbox infrastructure to host handheld and pre-release devices for driver build deployment and automated game benchmarking using Azure DevOps, Postgres SQL, Powershell, and Python
- Owned the Xbox DirectX runtime validation and implemented tooling to visualize execution timing in C++
- Enabled high frame rate support and HDR plane composition for the Xbox consoles in the kernel mode driver
- Instrumented the Xbox user mode driver to capture GPU command buffers and repeatedly resubmit work to track execution timing within nanoseconds across driver and shader compiler changes in C++
- Doubled mesh shader throughput by implementing a mixed-precision multiplication in GPU microcode
- Owned the GPU Pipeline View and implemented debug visualizations for alpha transparencies, color formatting, and texture resources for the GPU Performance Investigator for Xbox in HLSL, C++, and C#
- Mentored team interns and intern cohort career development workshops in Q&A sessions and morale events

Medtronic
Northridge, CA

Software Engineer Intern May 2019 - August 2019

- Developed automated testing framework in Java for the Minimed insulin pump and integrated the framework with the company wide testing platform hosted in Microsoft Azure. Expanded the framework to showcase real time unit testing results and metadata visualization for Android/iOS mobile app crash reporting.

Johns Hopkins Computer Science Department
Baltimore, MD

Head Teaching Assistant January 2019 - May 2020

- Implemented autograder support for grading student assignments in Python
- Lead team of course assistants to set learning goals for introductory programming course

University of Virginia
Charlottesville, VA

Systems Bioengineering Research Intern May 2018 - August 2018

- Analyzed terabytes of human genomic data in R to correlate inflammatory cell response with coronary artery diseases
- Developed R package and C++ workflows for filtering and sorting gene expression types
- Accepted to present at 2018 BMES conference in Atlanta, GA and 2018 REUS conference in Alexandria, VA

Technical Skills

Programming Languages Python, C/C++, Java, C#, R

Tools/Frameworks Git, DirectX, *nix, SQL, Windbg, Matlab, JUnit

Education

Johns Hopkins University
Baltimore, MD

B.S. BIOMEDICAL ENGINEERING, B.S. COMPUTER SCIENCE August 2016 - May 2020

- Honors: CS Honors Graduate, Dean's List, UPE Honors Society, Hopkins Academic Success Award